Higher National Diploma in Information Technology

First Year, Second Semester Examination – 2023

HNDIT2042-Data Communication and Computer Networks

Instructions for Candidates:
Answer five (5) questions only
Non programmable calculators are allowed

No. of questions : 6
No. of pages : 4
Time : Three (3) hours

Question 01 **[Total 20 Marks]**

i) What is communication Network? (2 Marks)

ii) What are the three Components of client server architecture? (3 Marks)

iii) Compare and contrast mail servers and web servers based on their function and usage. (4 Marks)

iv) Name any two types of network cables commonly used in networking. (2 Marks)

v) State two advantage of using fiber optic cables over copper cables. (4 Marks)

vi) What are the color codes used in a standard Ethernet (Cat 5e) cable following the T568B wiring scheme? (5 Marks)

Question 02 **[Total 20 Marks]**

i.) What is leased line in computer network? (2 Marks)

ii.) List two advantages of using a leased line for a business. (4 Marks)

iii.) A bank uses a leased line to connect its branches. Explain why this is important for the Bank's operations. (4 Marks)

iv) List any four applications of IoT (4 Marks)

v) What is the role of cloud computing in IoT? (4 Marks)

vi) Describe two situations where wireless media would be the most suitable choice for setting up a computer network. (2 Marks)

Question 03 **[Total 20Marks]**

i.) Give two functions of the Data Link Layer? (2 Marks)

ii.) Which OSI layer is responsible for encryption and compression? (2 Marks)

iii.) Name any two protocols used at the Transport Layer. (2 Marks)

iv.) Give the protocol used in the following scenario

a.) Perera is downloading a large PDF file from a university website.

b.) Minugi plays a fast-paced online game

c.) Saman connects her phone to a public Wi-Fi hotspot and received an IP address

(6 Marks)

v.) Explain three drawbacks of using Social Media (3 Marks)

vi.) "A junior technician installs a switch in place of a router to connect a LAN to the internet. Why won't this work, and how would you explain the difference between the two devices?" (5 Marks)

Question 04 **[Total 20 Marks]**

i) What is the main difference between IPv4 and IPv6? (2 Marks)

ii) What is the typical prefix length for an IPv6 subnet? (2 Marks)

iii) a.) Simplify the following IPv6 address:

2001:0db8:0000:0000:0000:ff00:0042:8329 (2 Marks)

fe80:0000:0000:0000:0202:b3ff:fe1e:8329 (2 Marks)

b) Expand the following compressed IPv6 address: **::1** (2 Marks)

iv.) You are provided with the IPv4 network 192.168.10.0/24. Your task is to subnet this network into four equal parts.

a) What will be the new subnet mask? (2 Marks)

b) What are the network addresses of all four subnets? (4 Marks)

c) What is the broadcast address of the second subnet? (2 Marks)

d) What is the range of usable host addresses in the third subnet? (2Marks)

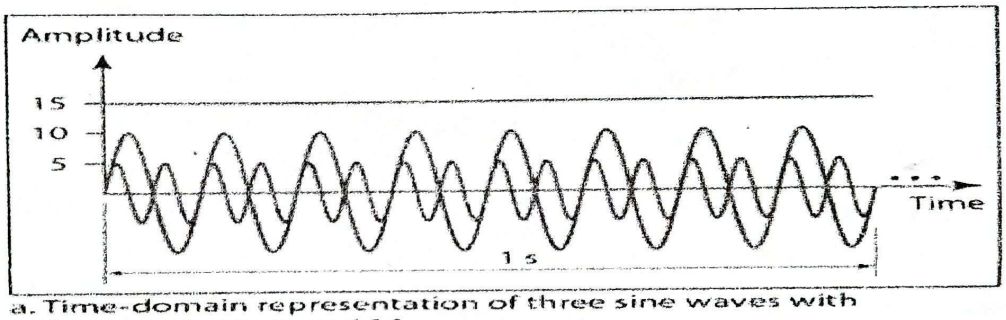
Question 05 **[Total 20 Marks]**

i.) Compare analog and digital signals. (4 Marks)

ii) Draw a time domain plot of an analog signal with the frequency of 6Hz and amplitude of 4 (4 Marks)

iii.) What is the frequency and amplitude of the Analog Signal $5\sin 200\pi t$, where t is the time (4 Marks)

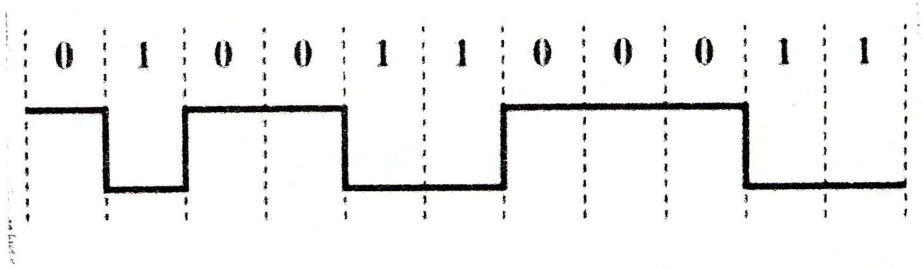
iv.) Draw a frequency domain plot for the time domain of two sine waves and DC signal as shown in **Figure a.** below. (3 Marks)



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16

v.) Consider the digital signal below (Non return to Zero-Level) and draw a Non return to Zero Inverted diagram for it. (5 Marks)

NRZ-L



Question 06 **[Total 20 Marks]**

i.) What is Rule-Based Access Control (RuBAC)? (3 marks)

ii.) How does RuBAC differ from Role-Based Access Control (RBAC)? (4 Marks)

iii.) State three Computer Security Objective (3 Marks)

iv.) Gives three Examples of Business functions and possible risks (4 Marks)

v.) Your computer running very slowly and you notice many pop-up ads even when you're not browsing the internet. What might be the cause, and how should you respond to protect your system? (6Marks)